

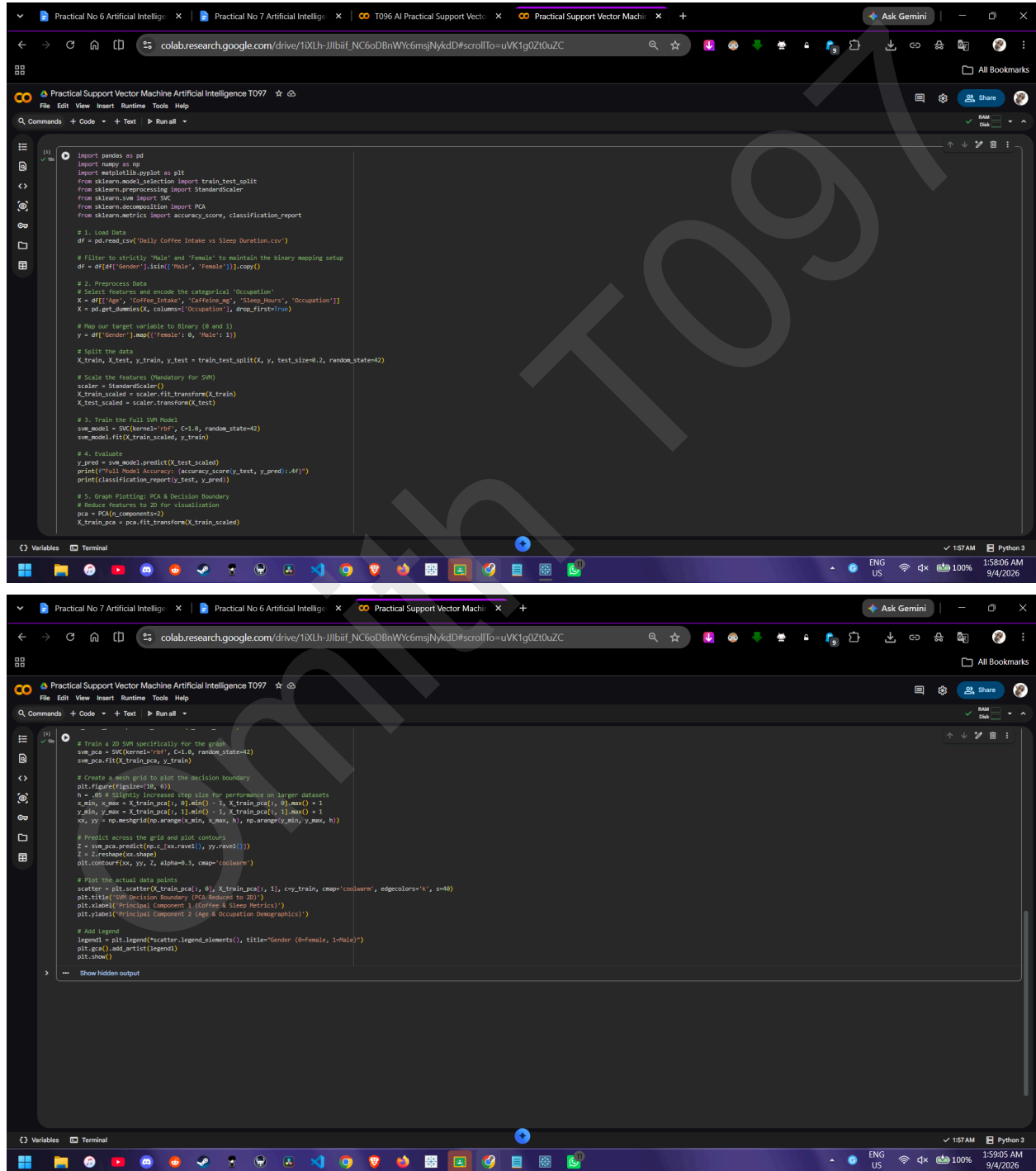
SHETH L.U.J & SIR M.V COLLEGE
Data Visualization with Power BI and Tableau

Practical No 5

Aim: Support Vector Machine

- Implement the SVM algorithm for binary classification.
- Train an SVM model using a given dataset and optimize its parameters.
- Evaluate the performance of the SVM model on test data and analyze the results.

Code:



The image displays two screenshots of a Google Colab notebook titled 'Practical Support Vector Machine Artificial Intelligence T097'. The first screenshot shows the initial data loading and preprocessing steps, including importing libraries, loading the 'Daily Coffee Intake vs Sleep Duration.csv' dataset, filtering for 'Male' and 'Female' genders, preprocessing categorical features like 'Age', 'Coffee Intake', 'Caffeine mg', 'Sleep Hours', and 'Occupation', and splitting the data into training and testing sets. The second screenshot continues with training an SVM model using a radial basis function kernel, evaluating its performance, and plotting the decision boundary and data points using PCA for visualization. The code is written in Python and includes comments for each step.

```
[1] In [ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.decomposition import PCA
from sklearn.metrics import accuracy_score, classification_report

# 1. Load Data
df = pd.read_csv('Daily Coffee Intake vs Sleep Duration.csv')

# Filter to strictly 'Male' and 'Female' to maintain the binary mapping setup
df = df[df['gender'].isin(['Male', 'Female'])].copy()

# 2. Preprocess Data
# Select features and encode the categorical 'Occupation'
X = df[['Age', 'Coffee Intake', 'Caffeine mg', 'Sleep Hours', 'Occupation']]
X = pd.get_dummies(X, columns=['Occupation'], drop_first=True)

# Map our target variable to Binary (0 and 1)
y = df['gender'].map({'Female': 0, 'Male': 1})

# Split the data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Scale the features (Mandatory for SVM)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# 3. Train the Full SVM Model
svm_model = SVC(kernel='rbf', C=1.0, random_state=42)
svm_model.fit(X_train_scaled, y_train)

# 4. Evaluate
y_pred = svm_model.predict(X_test_scaled)
print("Full Model Accuracy: (accuracy_score(y_test, y_pred):.4f)")
print(classification_report(y_test, y_pred))

# 5. Graph Plotting: PCA & Decision Boundary
# Reduce features to 2D for visualization
pca = PCA(components=2)
X_train_pca = pca.fit_transform(X_train_scaled)
```

```
[1] In [ ]: # Train a 2D SVM specifically for the graph
svm_pca = SVC(kernel='rbf', C=1.0, random_state=42)
svm_pca.fit(X_train_pca, y_train)

# Create a mesh grid to plot the decision boundary
plt.figure(figsize=(10, 6))
h = .45 # Slightly increased step size for performance on larger datasets
x_min, x_max = X_train_pca[:, 0].min() - 1, X_train_pca[:, 0].max() + 1
y_min, y_max = X_train_pca[:, 1].min() - 1, X_train_pca[:, 1].max() + 1
xx, yy = np.meshgrid(np.arange(x_min, x_max, h), np.arange(y_min, y_max, h))

# Predict across the grid and plot contours
Z = svm_pca.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)
plt.contourf(xx, yy, Z, alpha=0.3, cmap='coolwarm')

# Plot the actual data points
scatter = plt.scatter(X_train_pca[:, 0], X_train_pca[:, 1], c=y_train, cmap='coolwarm', edgecolors='k', s=40)
plt.title('SVM Decision Boundary (PCA Reduced to 2D)')
plt.xlabel('Principal Component 1 (Coffee & Sleep Metrics)')
plt.ylabel('Principal Component 2 (Age & Occupation Demographics)')

# Add Legend
legend = plt.legend(scatter.legend_elements(), title='Gender (0=Female, 1=Male)')
plt.gca().add_artist(legend)
plt.show()
```

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Output:

